

SABAL KC

Full-Stack / Backend Software Engineer | Go, Javascript, Node.js, Python, React, AWS

Houston, TX | (281) 905-3660 | sabalkc462@gmail.com | linkedin.com/in/sabal-kc | github.com/sabal-kc | sabalkc.com.np

Experience

Research Assistant (Software Developer)

September 2024 – Present

University of Houston, Office of Institutional Research

Houston, Texas

- Developed a full-stack centralized reporting platform using **HTML, CSS, JavaScript and PHP** to automate annual assessment reporting for 200+ degree programs.
- Modernized reporting infrastructure by building a KPI extraction pipeline in **SQL and PHP**, replacing manual reporting of metrics and ensuring 100% data accuracy from Oracle tables in Data Warehouse.

Software Engineer

May 2021 – July 2024

Dapi (YC W20)

United States (Remote)

- Engineered and maintained backend services for an open-banking ecosystem using **Go (GoKit, Fiber) and MongoDB**, utilizing **NATS** and Protocol Buffers for low-latency service communication.
- Created and scaled an in-house browser-as-a-service system using **Chromium, Node.js, Docker, and Kubernetes**, achieving 90% cost reduction by eliminating third-party services.
- Led a team of 4 engineers to develop and maintain web scraping and automation scripts in **TypeScript, Puppeteer and Node.js** for open banking integration across 30+ banks in the UAE and US, achieving a 95% success rate.
- Managed Dapi client libraries and SDKs in **Python, PHP, NodeJS** and maintained comprehensive documentation, reducing client integration time.
- Implemented new frontend features for the **React**-based client dashboard for sandbox features and API analytics, providing real-time visibility into system usage.
- Built and managed CI/CD pipelines using GitHub Actions and Porter for automated deployments.

Full-Stack Developer Intern

December 2020 – April 2021

Okhati Solutions (Healthcare Startup)

Kathmandu, Nepal

- Developed and maintained an EMR software used by clinics, labs, and hospitals, leveraging **React, Redux, Node.js with Express, and PostgreSQL** to streamline clinical and laboratory data management.
- Led the integration work for government COVID-19 data reporting, enabling tracking of 1000+ users daily.

Technical Skills

Programming languages: Go (Golang), JavaScript, TypeScript, Python, PHP

Backend & APIs: Node.js, Express, Flask, Supabase, Microservices, REST APIs

Databases: Oracle, MongoDB, PostgreSQL

Cloud & Containerization: AWS (EC2, S3, ECS, Fargate, SageMaker), Docker, Kubernetes, CI/CD

Developer Tooling & Automation: Git, Web scraping and browser automation (Puppeteer, Playwright), Postman

Education

University of Houston

August 2024 – May 2026

M.S. in Computer Science

Houston, Texas

Pulchowk campus, IOE, Tribhuvan University

November 2016 – April 2021

B.E. in Computer Engineering

Lalitpur, Nepal

Projects

Moodle/Canvas AI Agent, LangGraph, Python, Playwright

- Built a LangGraph agent that automates LMS login, assignment discovery, and assignment submission via Playwright and an agentic tool loop.
- Deployed as a containerized long-running service on AWS with scheduled execution and logging through LangSmith.

AWS Cloud-Based Movie Recommendation System, Go, AWS, React

- Built a scalable personalized movie recommender using AWS SageMaker ML models and OpenAI API with a Go backend and React frontend.
- Containerized backend services with Docker and deployed on AWS ECS (Fargate and EC2), comparing cost and performance trade-offs.